# Class Bot — User Guide

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How to run the Discord class server every day: attendance, announcements, students,
Excel files, updates and fixing problems.


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## 1. The big picture

The bot runs **24/7 on the Tinker Board**. You control it from Discord, in the private channel `🛠️|admin`, which only you can see. Code lives on GitHub; you edit it on your PC.

```
PC (edit)  ──git push──▶  GitHub  ──auto, within 5 min──▶  Tinker Board (bot runs here)
```

| Server part | What it is for |
|---|---|
| 👋 WELCOME | New students only. Section buttons to register. Hidden after they verify. |
| 🏫 ALL STUDENTS | 📌 notice-board (everyone) and 🔊 lounge. |
| 🌐 Course groups | One per course, shared by its sections (e.g. CSE 2109/2110 · OOP II · 3D, 3E, 3F, 3G). |
| 📘 Sections (3D, 3E, 3F, 3G, 4B, 4P) | 📢 announcement · 💬 general-text ✍️ · 📚 resources · 📊 marks-sheet · 🎞️ presentation-slide ✍️ · 🗓️ routine · 🧪 lab (OOP II only) · 🔊 voice channel |
| 🔒 ADMIN | 🛠️ admin (panel + commands) and 📜 bot-logs (automatic logs, daily backup). |

> ✍️ Students can write, attach files and create threads **only** in **general-text** and **presentation-slide**. Everywhere else they can read and react.

## 2. How students join

1. You click 🔗 **Invite link** in the panel and post the ready message in the Facebook / WhatsApp class group.
2. The student opens the link, goes to `👋|welcome` and clicks **their section button**.

3. A form opens: **Student ID** (11 digits) and **full name** → Submit.
4. Done. They get their section role, the nickname `1964_3D_Rahim Uddin`, a "👉 Start here" link to their section's pinned welcome card, and the welcome area disappears.

Backup way: a student can type `3D 11250121964 Rahim Uddin` in `🪪|id-name-section`. If a student leaves and comes back, their section is restored automatically.

> 🟡 If someone uses a Student ID that another account already used, they are **not** verified automatically. You get an **Approve / Reject** message in the admin channel.

## 3. The control panel

The panel is a set of messages in `🛠️|admin`: **each row is one message** — it lists what every button does, and **the buttons are right under that text**. Click a button → a **private menu** opens (only you see it) → choose from dropdowns or fill a short form → the result is posted in the admin channel.

> 🎛️ **Every bot message** in the admin channel ends with two small buttons: 🎛️ **Panel** (opens the whole panel privately, right there — no scrolling) and 📖 **All commands** (every text command with an example).

**Row 1 · Communication**

📢 Announce | ✉️ DM students | 🗓️ Scheduled posts | 🔗 Invite link | 📎 Marks / roster

**Row 2 · Attendance**

▶️ Start | ⏹️ Stop | ⏱️ Running now | 📈 Report | 📊 Stats

**Row 3 · Students**

📋 Student list | 📗 Excel | 🔎 Find | 🟡 Pending | 🔔 Remind unverified

**Row 4 · Manage members**

✅ Verify | 🔀 Move section | ✏️ Rename | ❌ Unverify | 🚪 Kick unverified

**Row 5 · System**

📌 Re-post welcome | 💾 Backup | 🖥️ Status | 🔄 Update | ♻️ Restart

**Row 6 · Routine & reminders**

🔄 Update routine | 🗓️ Show routine | 📅 My classes today | 📆 My week | ⏰ Remind a message

**Row 7 · Help**

📖 All commands | 🎛️ Show panel here

Panel missing or old? Type `!panel` in the admin channel to post it again.

## 4. Attendance

### Take attendance in a class

1. Click ▶️ **Start attendance**.

2. Choose **section** → **course** (for lab sections choose theory or lab) → **auto-stop time** (5–120 min, or no auto-stop) → **Start**.
3. The bot posts "Attendance is open" in that section's announcement channel and pings the students.
4. Students join the section's 🔊 **voice channel**. The bot records when each student joins and leaves.
5. Click ⏹️ **Stop attendance** (or let auto-stop do it). An **Excel sheet** is posted.

### What the Excel sheet shows

| Top part | Course, section, date, start time, stop time, **class duration**, taken by, present count |
|---|---|
| **Each student** | Status (**P** / **A**), first join, last leave, **minutes in class**, % of class time, number of joins |

Leaving and joining again is added correctly. Students of another section who joined are listed as `P*`.

### Semester report

Click 📈 **Report** → section → course. You get an Excel file with one column per class (`P (42m)` or `A`), total present and %. Students **below 60%** are marked red.

> 💡 Want a minimum stay? In `config.json` set `"attendance_min_percent": 50`. Then a student must stay at least half the class to be counted present.

Commands: `!att_start 3D 15` · `!att_start 3D CSE2110 30` · `!att_stop 3D` · `!att_status` · `!att_report 3D CSE2109`

## 5. Announcements & messages

### Post an announcement

1. Click 📢 **Announce**.
2. Choose one or more targets: All students, a section, a section's lab channel, or a course (all its sections).
3. Click **Write & ping** (students get a notification) or **Write (no ping)**.
4. Type a title and the message → Submit.

### Post later (scheduled)

In the same form, fill the **Schedule** box: `2026-10-26 09:00`, `21:30` (today or tomorrow) or `+2h` / `+30m`. See or cancel waiting posts with 🗓️ **Scheduled posts**.

### Announcement with a file (PDF, slide, image)

Buttons can't carry files, so type the command and attach the file to the same message:

```
!announce 3D Lecture 5 slides are attached
```

### Private DM to students

Click ✉️ **DM students** → choose targets → write the message. Every student gets it privately (about 1 second per student).

## 6. Students & Excel

| I want to... | Do this |
|---|---|
| See how many students | 📊 **Stats** |
| See a section's students | 📋 **Student list** → section |
| Get the list in Excel | 📗 **Excel** → Everything / a section / a course. One sheet per course + section, plus a Summary sheet. |
| Find one student | 🔎 **Find** → ID, name or Discord name |
| Remind people who didn't verify | 🔔 **Remind unverified** → they get a DM and a mention |
| Check who hasn't joined at all | Upload the official list once: `!roster_upload 3D` + attach CSV (ID, Name). Then `!missing 3D`. |

## 7. Fixing a student

| Problem | Button | What happens |
|---|---|---|
| Can't verify / wrong info | ✅ **Verify** → member + section → ID + name | Verified by you |
| Chose the wrong section | 🔀 **Move section** → member + section | Roles and nickname change |
| Name spelled wrong | ✏️ **Rename** → member → new name | Nickname fixed |
| Not your student | ❌ **Unverify** → member | Access and record removed |
| Same ID used twice | 🟡 **Pending** → Approve / Reject | You decide |
| Joined but never verified | 🚪 **Kick unverified** → days → Show list → Kick | They are removed |

## 8. Marks by private DM

Each student receives **only their own marks**. Nobody else can see them.

1. Make an Excel or CSV file. First row = column names. One column must be the Student ID.
2. In `🛠️┃admin` type the command and attach the file to the same message:

```
!marks_dm CT-1 Marks
```

| Student ID | Name | CT-1 (10) | Attendance (5) | Total |
|---|---|---|---|---|
| 11250121964 | Rahim Uddin | 8 | 5 | 13 |
| 11250121965 | Karima Akter | 9 | 4 | 13 |

The bot reports how many were sent and which IDs are not registered on the server.

## 9. Changing settings & updating

Settings are in `config.json` (on your PC, in the project folder). Common changes:

| Change | Where in config.json | After pushing |
|---|---|---|
| Your info, email, WhatsApp | `"instructor"` | nothing (automatic) |
| Invite link | `"invite_link"` (empty = bot makes one) | nothing (automatic) |
| Channel names / icons | `"channels"` | `!setup_server` |
| Add / remove a section or course | `"sections"`, `"courses"` | `!setup_server` |
| Attendance minimum % | `"attendance_min_percent"` | nothing |

### How to send a change to the Tinker Board

Open the project folder → click the address bar → type `cmd` → Enter. Then:

```
git add .
git commit -m "what I changed"
git push
```

Within **5 minutes** the bot pulls the change and restarts by itself (a message appears in 📜 bot-logs). In a hurry? Click 🔄 **Update**. If the new code or config has a mistake, the bot **does not** restart and tells you the error.

> 🛑 **Never** type `!reset_server` unless you really want to delete **every** channel and message. Both `!setup_server` and `!reset_server` ask **Yes / Cancel** first — read the warning.

### New semester

1. Edit `semester`, `courses` and `sections` in config.json.
2. git push → wait for the update → type `!setup_server` → Yes.
3. Click 🔗 **Invite link** and share it with the new students.

## 10. Tinker Board care

From your PC's cmd: `ssh shovon@192.168.0.221`, then:

| Command | What it does |
|---|---|
| `bash ~/discord-bot/scripts/bot.sh status` | Is the bot running? |
| `bash ~/discord-bot/scripts/bot.sh log` | Live log (Ctrl+C to exit) |
| `bash ~/discord-bot/scripts/bot.sh restart` | Restart the bot |

| Command | What it does |
| --- | --- |
| `bash ~/discord-bot/scripts/bot.sh update` | Pull from GitHub + restart (if !update fails) |
| `sudo systemctl stop nubtk-bot` | Stop the bot |

⚠️ **Run the bot in one place only.** If it runs on the PC and the Tinker Board at the same time, everything happens twice and the data splits.
To move it to the PC: stop it on the board, then copy the data: `scp -r shovon@192.168.0.221:~/discord-bot/data .`

💾 A backup zip of all data is posted in 📜 bot-logs every night at 23:30. Click 💾 **Backup** any time for a fresh one. Student data never goes to GitHub.

## 11. All commands (with examples)

Type these in `🛠️┃admin`. `!help` shows the same list in Discord.

| Command | What it does | Example |
|---|---|---|
| **🏗️ Setup** | | |
| `!panel` | Re-post the button control panel | `!panel` |
| `!setup_server` | Create / sync all channels & roles (asks to confirm) | `!setup_server` |
| `!setup_welcome` | Re-post welcome buttons + instructor cards | `!setup_welcome` |
| `!invite` | Invite link + join guide for students | `!invite` |
| `!reset_server` | ⚠️ Delete EVERYTHING, build fresh (asks to confirm) | `!reset_server` |
| **🎓 Students** | | |
| `!stats` | Count per section & course | `!stats` |
| `!list <sec>` | All students of a section | `!list 3D` |
| `!find <id/name>` | Search a student | `!find 11250121964 · !find Rahim` |
| `!verify @user <sec> <id> <name>` | Verify manually | `!verify @rahim 3D 11250121964 Rahim Uddin` |
| `!unverify @user` | Remove access + record | `!unverify @rahim` |
| `!move @user <sec>` | Change section | `!move @rahim 3F` |
| `!rename @user <name>` | Fix name / nickname | `!rename @rahim Md Rahim Uddin` |
| `!pending` | Requests waiting for approval | `!pending` |
| `!notify_unverified` | Remind unverified members | `!notify_unverified` |
| `!kick_unverified <days> [confirm]` | List / kick never-verified | `!kick_unverified 7` → `!kick_unverified 7 confirm` |
| `!roster_upload <sec>` | Upload official list (attach CSV) | `!roster_upload 3D` + attach `3D.csv` |
| `!missing <sec>` | Roster students not joined yet | `!missing 3D` |
| **📢 Communication** | | |
| `!announce <target> <text>` | Post + ping (all / 3D / 3D:lab / CSE2109) | `!announce 3D Class test on Sunday at 10 AM` |
| `!announce_silent <target> <text>` | Same, no ping | `!announce_silent all Slides are uploaded` |

| Command | What it does | Example |
|---|---|---|
| `!dm <target> <text>` | DM every student | `!dm 4P Tomorrow's class is online` |
| `!marks_dm <title>` | Send each student their own marks (attach CSV/Excel) | `!marks_dm CT-1 Marks + attach ct1.xlsx` |
| `!scheduled` | List / cancel scheduled announcements | `!scheduled` |
| **🗓️ Routine & reminders** | | |
| `!today` | My classes today: section, course, time, room | `!today` |
| `!tomorrow` | My classes tomorrow | `!tomorrow` |
| `!week` | My whole weekly routine | `!week` |
| `!routine_check` | Update routine: check the online sheet, update pinned routines, announce changes | `!routine_check` |
| `!routine [sec] [all\|mine]` | Show saved routine: full section / only my classes | `!routine 3D · !routine 3D mine` |
| `!routine_repost` | Re-post every section's pinned routine | `!routine_repost` |
| (right-click a message → Apps → ⏰ Set reminder) | Re-announce a message now or later | `deadline / CT reminder` |
| **🗓️ Attendance** | | |
| `!att_start <sec> [course] [minutes]` | Start voice attendance | `!att_start 3D 15 · !att_start 3E CSE2110 30` |
| `!att_stop <sec>` | Stop + get CSV | `!att_stop 3D` |
| `!att_status` | Running sessions | `!att_status` |
| `!att_report <sec> [course]` | Semester % per student | `!att_report 3D CSE2109` |
| **💾 Data & System** | | |
| `!excel [target]` | Student list as Excel | `!excel · !excel 3D · !excel CSE2217` |
| `!export [target]` | Student CSV/JSON files | `!export CSE2109 3D` |
| `!backup` | Zip of all data | `!backup` |
| `!status` | Uptime, CPU temp, RAM | `!status` |
| `!logs [lines]` | Last log lines | `!logs 50` |
| `!update` | Pull from GitHub + restart | `!update` |
| `!restart` | Restart the bot | `!restart` |

## 12. Problems & fixes

| Problem | Fix |
|---|---|
| Bot is offline | Check the Tinker Board has power and internet. SSH in → `bot.sh status` → `bot.sh restart`. |
| Buttons say "interaction failed" | The bot was restarting. Wait 20 seconds and try again. Still failing → `!panel`. |
| Commands get no reply | Type them in `🛠️┃admin`, not in another channel. |
| Channels appear twice | The board did not have the `data` folder. Run `!reset_server` once → Yes. |
| Student says "I can't see my channels" | 🔎 **Find** them. Not verified → ✅ **Verify**. Wrong section → 🔀 **Move**. |
| Student didn't get a DM | Their Discord privacy blocks DMs from servers. Ask them to allow it (Server → Privacy Settings → Direct Messages). |
| Update didn't arrive | Check `git push` worked. Click 🔄 **Update** and read the message. |
| Bot can't give roles / change nicknames | Server Settings → Roles → drag the bot's role to the top. |

## 13. Routine & reminders

### Class routine (from the department Google Sheet)

1. When the routine changes, click 🔄 **Update routine** — in the panel (Row 6) **or right under the pinned routine** in any 🗓️┃routine channel. It also runs by itself every 6 hours.
2. The bot reads the official routine sheet for every section.
3. Each section's `🗓️┃routine` channel keeps **one pinned message = the latest routine**, with two parts:
   📚 **Full class routine** — all courses and teachers of the section (⭐ = your classes)
   👨‍🏫 **Classes with SM** — only your classes with day, time and room.
4. If something changed, the old pinned message is replaced and the bot posts what changed: ➕ new · ➖ removed, and pings the section.

> ⚠️ The Google Sheet must be shared as **Anyone with the link → Viewer**, otherwise the bot cannot read it. The link is in `config.json → routine.sheet_url`.

### My classes today

Click 📅 **My classes today** (Row 6) or type `!today`, `!tomorrow`, `!week`. You see every class with **time, section, course and room**. Every morning at **7:30 AM** the bot also posts today's classes in the admin channel (nothing on days without classes).

## Reminder for a message (deadline, CT...)

1. **Easiest:** right-click the message (e.g. the assignment announcement) → **Apps** → ⏰ **Set reminder**.
   Or click ⏰ **Remind a message** in the panel and paste the message link (right-click → Copy Message Link).
2. **When?** leave empty to post now, or write times: `2026-10-28 20:00, 2026-10-29 08:00` · `+1d` · `21:30`.
3. Optional note, e.g. "Assignment 2 deadline is tomorrow 11:59 PM!".
4. **Where?** empty = the same section or course as the message, or type `3D`, `CSE2109`, `all`.

Show it in the admin channel: `!routine 3D` (both) · `!routine 3D all` · `!routine 3D mine`.

The reminder is posted in the announcement channel with a ping, a quote of the message and an **Open the original message** button. Waiting reminders are listed under 🗓️ **Scheduled posts**, where you can cancel them.

No "Apps → ⏰ Set reminder" in the right-click menu? Open this link once and click Authorize (the bot stays in the server):
`discord.com/oauth2/authorize?client_id=1297622424640294922&permissions=8&scope=bot+applications.commands`

Code: github.com/shovon-mandal/academic-discord-management-bot · Secret token is only in the `.env` file — never share it or upload it.